

WEDO ROBOTICS



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES (1 HOUR PER
CLASS)**

**MODE:
ONLINE**

WeDo is an engaging robotics program designed to introduce young learners to the exciting world of **Science, Technology, Engineering, Arts, and Mathematics (STEAM)** through hands-on model building and visual programming. Students construct interactive robotic models, understand how real-world machines work, and bring their creations to life using **block-based coding**. Every session combines **mechanical engineering concepts, scientific exploration, computational thinking, and creative problem-solving**, making learning fun, meaningful, and practical. Throughout the course, students develop the confidence to design, build, test, improve, and program their own robotic solutions while strengthening logical thinking and collaboration skills.

COURSE CURRICULUM

LEVEL 1 – Robotics Foundations

Science Story– Students begin their robotics journey by exploring how simple machines and robotic systems work in everyday life. Through hands-on engineering and visual programming, they build a strong foundation in mechanics, coding, and problem-solving.

1. Introduction to Robotics & Smart Machines

- Understand the basics of robots, motors, and movement.
- Create simple robot actions using visual block programming.

2. Wheels, Axles & Motion

- Explore how wheels and axles create balanced movement.
- Program robots to move with speed and directional control.

3. Gears & Motion Transfer

- Learn how gears transfer force, speed, and motion.
- Build mechanisms that demonstrate gear-driven movement.

4. Simple Machines & Mechanical Linkages

- Discover levers, cranks, and linkage mechanisms.
- Program repetitive movements using loops and sequences.

5. Structures & Engineering Design

- Explore strong structures, stability, and load distribution.
- Apply engineering design principles through build challenges.

6. Transportation & Industrial Robotics

- Understand lifting systems and material handling in industries.
- Program robots to perform simple automated tasks.

7. Flight & Rotational Motion

- Learn the science of lift, thrust, and rotational motion.
- Control motor speed to simulate flying mechanisms.

8. Environmental Robotics

- Discover how robotics supports recycling and sustainability.
- Create automated solutions for environmental challenges.

9. Agricultural Robotics

- Explore robotics used in farming and ecosystem management.
- Program repetitive actions for smart agricultural systems.

10. Sensors & Smart Automation

- Learn how robots detect and respond to their surroundings.
- Program basic sensor-controlled robotic behaviors.

11. Industrial Automation

- Understand robotic arms and precision engineering.
- Program multi-step movements for automated operations.

12. Engineering Challenge & Innovation

- Apply mechanical and coding concepts to solve a design challenge.
- Develop creativity, teamwork, and problem-solving skills.

LEVEL 2 – Advanced Robotics & Automation

Science Story—Students advance into intelligent robotics by exploring autonomous systems, sensors, engineering design, and real-world technologies. They strengthen computational thinking while solving practical engineering challenges through robotics.

1. Space Robotics & Exploration

- Explore robotic systems used in planetary exploration.
- Program autonomous movement for challenging terrains.

2. Structural Engineering & Disaster Science

- Investigate earthquakes, stability, and structural strength.
- Design and test engineering solutions through experimentation.

3. Mechanical Advantage

- Understand pulleys and force multiplication in machines.
- Apply mechanical systems to lifting and climbing challenges.

4. Interactive Robotics

- Explore sound, music, and human-robot interaction.
- Program robots to respond through interactive behaviors.

5. Sensor-Based Robotics

- Learn how sensors enable intelligent decision-making.
- Program autonomous robots using conditional logic.

6. Precision Engineering

- Explore gears, lifting systems, and controlled movements.
- Program robots for accurate and efficient operations.

7. Speed, Motion & Performance

- Understand friction, gear ratios, and speed optimization.
- Compare robotic performance through practical investigations.

8. Automation & Computational Thinking

- Learn object detection, counting, and automated processes.
- Use variables and logical programming to solve challenges.

9. Biomimicry & Nature-Inspired Robotics

- Discover how nature inspires robotic movement and design.
- Program coordinated movement patterns based on living organisms.

10. Mathematics Through Robotics

- Explore patterns, measurements, and geometric designs.
- Apply mathematical thinking through creative robotics projects.

11. Design Thinking & Innovation

- Follow the engineering design process to solve real-world problems.
- Improve robotic solutions through testing and iteration.

12. Capstone Engineering Project

- Integrate mechanical design, coding, and STEAM concepts into a final project.
- Present an innovative robotic solution showcasing creativity and technical skills.

CLASS STRUCTURE

1) STEAM Concept Exploration

- Learn the scientific and engineering principles behind each robotic model through engaging discussions and real-world examples.

2) Mechanical Design & Construction

- Build functional robotic models using structural components, gears, wheels, axles, pulleys, motors, and sensors while understanding how each mechanism works.

3) Robotics Programming

- Program robotic models using an intuitive drag-and-drop coding environment to control movement, sensors, sounds, lights, loops, and decision-making.

4) Engineering Challenge & Innovation

- Test, modify, and improve robotic models through hands-on challenges, encouraging creativity, computational thinking, problem-solving, and teamwork.